

Exam Date & Time: 02-Dec-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

### Introduction to Photonics [NPT-3209]

Marks: 50

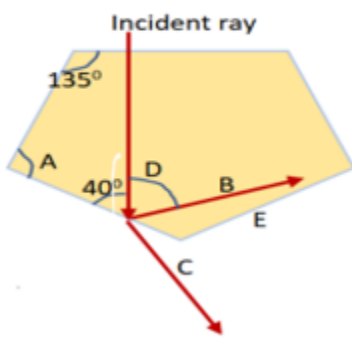
Duration: 180 mins.

#### PART-A

Answer all the questions.

- 1)
 

From the given figure What is the angle D in degrees?

A)  (3)

B) The intensity of the two coherent waves is 10. What is the resultant intensity if the two waves undergo constructive and destructive interference respectively? (3)

C) With a neat diagram discriminate colours using young's double slit experiment. ( $\lambda_{\text{red}} = 650\text{nm}$  &  $\lambda_{\text{orange}} = 600\text{nm}$ ) (4)
- 2)
 

Consider a 60W light bulb. Assume that most of the radiation given off is in the form of light of wavelength 650nm. What should be the number of photons given off per second. (3)

A)

B) A step index multimode fibre has a core of refractive index 1.5 and a cladding of refractive index 1.489. Calculate the temporal broadening of a pulse for a length of 1km of straight fibre in ns/km. (3)

C) Analyse light generation and amplification with the principles of photon interaction with atoms. (4)
- 3)
 

Consider an optical fibre of diameter 50 $\mu\text{m}$ . If the core is of refractive index 1.6, find the maximum number of total internal reflections that a ray incident at an angle 30 degree at the core cladding boundary will suffer in travelling through 1m of the straight fibre. (3)

A)

B) A 100 W sodium lamp emits light at wavelength of 600 nm. Assuming it to be 25% (3)

efficient in converting electrical energy to light. Calculate the number of photons the light emits per second (photon flux) in (photons/sec).

- C) With a neat block diagram explain optical fiber communication system. (4)
- 4) A Nd:YAG laser rod is composed of Nd ions in the form of impurities in a crystal of yttrium aluminum garnet host, that corresponds to a Nd<sup>3+</sup> ion density in the laser rod of about  $1.38 \times 10^{26} \text{ m}^{-3}$ . Suppose all these ions are pumped to their upper levels essentially all at once. From there they transit downward, emitting radiation at 1060 nm. (3)
- A) Determine the energy radiated by a) One photon b) the total amount of energy radiated per cubic meter of the rod
- B) A linearly polarized light is incident on a 2 cm long Terbium Aluminum Garnet (TAG) crystal with verdet constant of -41 rad/T-m. Calculate the magnitude of magnetic flux density which required to rotate the incoming polarization by an angle of 30° in Tesla. (3)
- C) Consider an InGaAs semiconductor laser ( $n=3.7$ ) emitting 1.3 μm wavelength consists of a 0.4 mm long cavity with internal losses of 40 cm<sup>-1</sup> and mirror reflectivity of 0.9 at either end. (a) Find the frequency spacing between the longitudinal modes of the laser. (4)
- (b) Find the minimum gain required to compensate the losses in the laser cavity.
- 5) Calculate the fiber length required to get the nonlinear phase shift induced by self-phase modulation of  $\pi/2$  in km. Assume zero fiber losses. (3)
- A)
- B) Assuming uniform excitation along the length, calculate the threshold gain (G). (3)
- C) Obtain the expression for stimulated Brillouin scattering. (4)

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